BRAC University

Department of Mathematics and Natural Sciences

Lecture on

Real Analysis (MAT221)

Riemann Integral

Upper and Lower Sums, Riemann Integrability

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Conducted by

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How Should Integration Be Defined?

In elementary calculus, integration is often introduced as the inverse operation of differentiation. For sufficiently nice functions, this relationship is expressed by the Fundamental Theorem of Calculus.

However, the idea of area should not depend on our ability to find an antiderivative.

The purpose of this chapter is to define integration directly from properties of the function itself.

The basic idea will be:

Approximate the area using rectangles and make the approximation arbitrarily accurate.

The Fundamental Theorem of Calculus

Fundamental Theorem of Calculus

Let $f : [a, b] \to \mathbb{R}$ be continuous. If F is any antiderivative of f on $[a, b]$, that is,

$$F'(x) = f(x),$$

then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Problem

Let

$$f(x) = 2x + 1$$

Use the Fundamental Theorem of Calculus to evaluate

$$\int_0^2 f(x)\,dx.$$

First find an antiderivative F of f, and then compute

$$F(2) - F(0).$$

Why Antiderivatives Are Not Enough

Consider the function

$$h(x) = \begin{cases} 1, & 0 \le x < 1, \\ 2, & 1 \le x \le 2. \end{cases}$$

Problem

Show that h has no antiderivative on $[0,2]$.

Nevertheless, the geometric area under the graph is easy to determine:

$$1 \cdot 1 + 2 \cdot 1 = 3.$$

Therefore, it is natural to expect that

$$\int_0^2 h(x)\,dx = 3.$$

Thus,

integration should be defined independently of antidifferentiation.

From Geometry to Approximation

Suppose that f is a nonnegative function defined on $[a, b]$.

We want to give a precise meaning to the phrase

the area under the graph of f from a to b.

The main geometric idea is to divide the interval into smaller pieces and approximate the region using rectangles.

Some rectangles will underestimate the area, while others will overestimate it.

If these two types of approximations can be made arbitrarily close, then there is only one possible value for the area.

Bounded Functions on Closed Intervals

Bounded Function

A function

$$f : [a, b] \to \mathbb{R}$$

is called **bounded** if there exists $M > 0$ such that

$$|f(x)| \leq M$$

for every $x \in [a, b]$.

Boundedness is essential in the construction of upper and lower sums.

If a function is unbounded on a subinterval, then a finite upper rectangular approximation may not exist.

Theorem

Every continuous function on a closed interval $[a, b]$ is bounded.

Partitions of an Interval

Partition

A **partition** P of $[a, b]$ is a finite ordered set

$$P = \{x_0, x_1, \ldots, x_n\}$$

such that

$$a = x_0 < x_1 < \cdots < x_n = b.$$

Norm of a Partition

Norm (Mesh)

Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$. The **norm**, or **mesh**, of P is

$$|P| = \max_{1 \le i \le n} (x_i - x_{i-1}).$$

Problem

Let $P = \left\{0, \frac{1}{4}, \frac{1}{2}, 1, 2\right\}$.

1. Compute $|P|$.
2. Add the point $\frac{3}{2}$ to obtain the partition $Q = \left\{0, \frac{1}{4}, \frac{1}{2}, 1, \frac{3}{2}, 2\right\}$.
3. Compute $|Q|$.
4. Compare $|P|$ and $|Q|$.

Supremum and Infimum on Subintervals

Definition

Let f be bounded on $[a, b]$, and let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition. For each subinterval $[x_{i-1}, x_i]$, define

$$M_i = \sup \{f(x) : x \in [x_{i-1}, x_i]\}$$

and

$$m_i = \inf \{f(x) : x \in [x_{i-1}, x_i]\} .$$

Lower Sums

Lower Sum

Let f be bounded on $[a, b]$, and let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$.

The **lower sum** of f with respect to P is

$$L(f, P) = \sum_{i=1}^{n} m_i(x_i - x_{i-1}).$$

Upper Sums

Upper Sum

Let f be bounded on $[a, b]$, and let

$$P = \{x_0, x_1, \ldots, x_n\}$$

be a partition of $[a, b]$.

The **upper sum** of f with respect to P is

$$U(f, P) = \sum_{i=1}^{n} M_i(x_i - x_{i-1}).$$

Problem

Let

$$f(x) = x^3 - 4x^2 + 2$$

on $[0, 4]$, and let

$$P = \{0, 1, 2, 3, 4\}.$$

1. Compute $L(f, P)$.
2. Compute $U(f, P)$.

Lower Sums Are Below Upper Sums

Theorem

For every partition P of $[a, b]$,

$$L(f, P) \leq U(f, P).$$

Refinement of a Partition

Refinement

Let P and Q be partitions of $[a, b]$. We say that Q is a **refinement** of P if

$$P \subseteq Q.$$

Note: More points in a partition does not necessarily mean that is a refinement. The refinement must contain all the points of the original partition.

Refinement and Monotonicity

Theorem

If Q is a refinement of P, then

$$L(f,P) \leq L(f,Q)$$

and

$$U(f,Q) \leq U(f,P).$$

Problem

Let $f(x) = x^2$ on $[0, 4]$.

Consider

$$P = \{0, 2, 4\}$$

and

$$Q = \{0, 1, 2, 4\}.$$

1. Verify that Q is a refinement of P.
2. Compute $L(f, P)$ and $U(f, P)$.
3. Compute $L(f, Q)$ and $U(f, Q)$.
4. Verify that
$$L(f, P) \leq L(f, Q).$$
5. Verify that
$$U(f, Q) \leq U(f, P).$$
6. Compare
$$U(f, P) - L(f, P)$$
with
$$U(f, Q) - L(f, Q).$$

Comparing Two Arbitrary Partitions

Theorem

If P_1 and P_2 are any two partitions (not necessarily refinements) of $[a, b]$, then

$$L(f, P_1) \leq U(f, P_2).$$

Thus, no matter which partitions are chosen, every upper sum is greater than or equal to every lower sum.

Note: Two partitions do not need to be refinements of each other.

However, given two partitions P_1 and P_2, their union

$$P_1 \cup P_2$$

is a partition that refines both of them.

Problem

Let

$$f(x) = x^2$$

on $[0, 4]$, and consider

$$P_1 = \{0, 1, 2, 4\}$$

and

$$P_2 = \{0, 2, 3, 4\}.$$

Neither partition is a refinement of the other.

1. Compute $L(f, P_1)$.
2. Compute $U(f, P_2)$.
3. Verify that
$$L(f, P_1) \le U(f, P_2).$$
4. Find the common refinement
$$P_1 \cup P_2.$$
5. Explain the chain
$$L(f, P_1) \le L(f, P_1 \cup P_2) \le U(f, P_1 \cup P_2) \le U(f, P_2).$$

Upper Integral

Instead of choosing one particular partition, consider the collection of upper sums obtained from *all* partitions of $[a, b]$.

Each upper sum is an overestimate.

The smallest possible such overestimate is obtained by taking an infimum.

Upper Integral

Let f be bounded on $[a, b]$.
The **upper integral** of f over $[a, b]$ is

$$\overline{\int_a^b} f = \inf \{U(f, P) : P \text{ is a partition of } [a, b]\}.$$

Lower Integral

Each lower sum is an underestimate.

The largest possible such underestimate is obtained by taking a supremum.

Lower Integral

Let f be bounded on $[a, b]$.

The **lower integral** of f over $[a, b]$ is

$$\underline{\int_a^b} f = \sup \{L(f, P) : P \text{ is a partition of } [a, b]\} .$$

The lower integral is the best possible lower approximation, while the upper integral is the best possible upper approximation.

Comparing the Lower and Upper Integrals

Theorem

For every bounded function

$$f : [a, b] \to \mathbb{R},$$

we have

$$\underline{\int_a^b} f \leq \overline{\int_a^b} f.$$

Problem

Let

$$f(x) = x$$

on $[0, 1]$.

For each $n \in \mathbb{N}$, consider the uniform partition

$$P_n = \left\{0, \frac{1}{n}, \frac{2}{n}, \ldots, \frac{n-1}{n}, 1\right\}.$$

1. Show that

$$L(f, P_n) = \frac{n-1}{2n}.$$

2. Show that

$$U(f, P_n) = \frac{n+1}{2n}.$$

3. Hence show that

$$U(f, P_n) - L(f, P_n) = \frac{1}{n}.$$

4. Determine

$$\lim_{n\to\infty} \left(U(f, P_n) - L(f, P_n)\right).$$

5. Determine

$$\lim_{n\to\infty} L(f, P_n) \qquad \text{and} \qquad \lim_{n\to\infty} U(f, P_n).$$

Riemann Integrability

The previous example suggests the essential idea of integration.

If lower sums and upper sums can be forced arbitrarily close together, then there should be a unique number representing the integral.

Riemann Integrable Function

A bounded function

$$f : [a, b] \to \mathbb{R}$$

is called **Riemann integrable** on $[a, b]$ if

$$\underline{\int_a^b} f = \overline{\int_a^b} f.$$

The common value is called the **Riemann integral** of f and is denoted by

$$\int_a^b f(x)\, dx.$$

In other words, f is Riemann integrable when there is no gap between the best lower approximation and the best upper approximation.

A Pathological Counterexample

Dirichlet Function

Define

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Both $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$.

Therefore, every nonempty interval contains both rational and irrational numbers.

Problem

Show that the Dirichlet function is not Riemann integrable on $[0, 1]$.

Thank You!

We'd love your questions and feedback.

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